Immunohistochemical expression of podoplanin (D2-40), lymphangiogenesis, and neoangiogenesis in tooth germ, ameloblastomas, and ameloblastic carcinomas.

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Ameloblastoma is a benign but locally aggressive odontogenic tumor, while ameloblastic carcinoma is its malignant counterpart. Angiogenesis and lymphangiogenesis in malignancies have been correlated with higher aggressiveness and poor prognosis, as well as greater expression of podoplanin by tumoral cells. Immunohistochemical expression of podoplanin, CD34, and CD105 (endoglin) was evaluated in 53 ameloblastomas and three ameloblastic carcinomas; additionally, immunohistochemistry for podoplanin was also performed in 10 tooth germs. Microvessel density of blood and lymphatic vessels was calculated and compared between ameloblastomas and ameloblastic carcinomas. Immunoexpression of podoplanin by ameloblastic cells was evaluated in tooth germs, ameloblastomas, and ameloblastic carcinomas. Podoplanin was similarly expressed by odontogenic epithelial cells of tooth germs and ameloblastomas, while its expression was lower in ameloblastic carcinomas. There was no difference in microvessel density assessed by CD34 between ameloblastomas and ameloblastic carcinomas; nevertheless, the latter presented higher amounts of lymphatic and new formed blood vessels. Results suggest that podoplanin does not seem to be involved in invasion mechanisms of ameloblastic carcinomas, as its expression was decreased in the malignant tumoral cells. On the other hand, the increased lymphatic microvessel density and neoangiogenesis found in ameloblastic carcinomas could be related to its aggressiveness and potential for metastasis.

FDG-PET findings of Ameloblastoma: a case report.

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Ameloblastoma is a benign odontogenic neoplasm of the jaw, rarely presenting as a malignant tumor. Although it is very important to discriminate ameloblastoma from ameloblastic carcinoma in order to decide the appropriate operative procedure, this is difficult using conventional CT and MRI. We report a case of maxillar ameloblastoma in a 78-year-old man where FDG-PET/CT was useful for making this discrimination. CT demonstrated a $31 \times 43 \times 46$-mm mass in the left posterior maxillary sinus with destruction of its posterior and lateral wall and alveolar bone. MRI demonstrated a hypo- to isointense heterogeneous pattern on T1WI, heterogeneous hyperintensity with a prominent high-signal spot on T2WI, high signal intensity on DWI reflecting restricted diffusion, and strong heterogeneous enhancement. Because FDG-PET/CT showed mild FDG uptake (SUVmax 2.40) by the mass, ameloblastoma, rather than ameloblastic carcinoma, was considered to be the correct diagnosis. It appears that ameloblastic carcinoma shows intense FDG uptake, whereas ameloblastoma shows mild or moderate FDG uptake, and only rarely intense FDG uptake. Our experience suggests that FDG-PET/CT may be effective for discriminating ameloblastoma from ameloblastic carcinoma. Especially, in cases showing mild FDG uptake, benign ameloblastoma would seem the most likely diagnosis. FDG-PET/CT may be useful as an adjunctive modality for diagnosis, treatment planning and surveillance of ameloblastoma and ameloblastic carcinoma.

Orosomucoid-1 Expression in Ameloblastoma Variants.

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Odontogenic tumors constitute a group of heterogeneous lesions of benign and malignant neoplasms with variable aggressiveness. Ameloblastomas are a group of benign but locally invasive neoplasms that occur in the jaws and are derived from epithelial elements of the tooth-forming apparatus. We previously described orosomucoid-1 protein expression in odontogenic myxomas. However, whether orosomucoid-1 is expressed in other odontogenic tumors remains unknown. Since orosomucoid-1 belongs to a group of acute-phase proteins and has many functions in health and disease, we identified and analyzed orosomucoid-1 expression in ameloblastoma variants and ameloblastic carcinoma using western blot and immunohistochemical techniques. Thirty cases of ameloblastoma were analyzed for orsomucoid-1; five specimens were fresh for western blot study (four benign ameloblastomas and one ameloblastic carcinoma), and 25 cases of benign ameloblastoma for immunohistochemical assays. Orosomucoid-1 was widely expressed in each tumor variant analyzed in this study, and differential orosomucoid-1 expression was observed between benign and malignant tumor. Orosomucoid-1 may play an important role in the behavior of ameloblastomas and influence the biology and development of the variants of this tumor.

Metastasizing Ameloblastoma: A 10 Year Clinicopathological Review with an Insight Into Pathogenesis.

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Ameloblastoma, a benign but locally aggressive odontogenic tumor, often demonstrates metastasis despite benign histological features and this variant is termed as metastasizing ameloblastoma (METAM). It was classified under the malignant category in the 2005 WHO but has been re-classified under benign epithelial odontogenic tumors in the latest 2017 WHO classification. The present review aimed at gathering the available data on METAM to update the current cognizance about the pathology. Comprehensive search of the databases (viz., PubMed, Medline, SCOPUS, Web of Science, EMBASE and Google Scholar) was done for published articles on METAM following Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. A total of 42 cases were extracted. The mean age of occurrence was 42.71 ± 15.87 years. A slight male predilection was noted. Mandibular cases showed more metastasis than maxillary cases. Follicular ameloblastoma was most frequently encountered at secondary site followed by plexiform type. Lungs were the most commonly affected secondary sites. METAM is a rare odontogenic tumor and the diagnosis is usually made in retrospect. Inadequate treatment may result in multiple recurrences and metastasis in rare instances. Metastasis in ameloblastoma appears to be multi-factorial in nature and needs further investigation in untapped territory like exploration of quantum effects at cellular and molecular levels.

Carcinoma Ex Ameloblastoma of the Mandible: A Rare Case Report.

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Ameloblastoma is a benign yet locally aggressive tumor of the jaw bones and is most commonly found in the lower mandibular region. Histologically, it shows benign characteristics. However, ameloblastoma can turn malignant to show a more aggressive clinical course. Carcinoma ex ameloblastoma is an extremely rare malignancy arising from a pre-existing long-standing ameloblastoma or a recurrence of an ameloblastoma. According to the literature search, six to seven cases have so far been documented, and the majority of the lesions had a propensity to metastasize. Here, we present a case of carcinoma ex ameloblastoma implicating a 19-year-old male patient manifesting in the mandible, which arises from pre-existing ameloblastoma.

Metastatic malignant ameloblastoma of the kidneys.

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Ameloblastoma is an uncommon disease in the urological field. The resulting tumors or cysts are of odontogenic epithelial origin, are usually benign in nature and rarely metastasize to distant organs. We describe a case of metastatic ameloblastic carcinoma in both kidneys of a 38-year-old Japanese man, who had a history of malignant ameloblastoma and was referred to us for evaluation because of gross hematuria and left flank pain. Computed tomography showed irregular cystic masses in both kidneys. After we confirmed that the primary lesion and the lung metastatic lesion had not recurred, we treated the patient surgically. Approximately 4 months postoperatively the patient suffered a local recurrence of tumors that was very invasive and aggressive. The patient died 2 months later and the autopsy showed local metastasis only, without any metastatic lesion in the lungs or other organs. The present case showed that malignant ameloblastoma is highly aggressive, and in the case of metastases the prognosis is usually extremely poor.

Hypoxia-induced HIF-1α and ZEB1 are critical for the malignant transformation of ameloblastoma via TGF-β-dependent EMT.

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Ameloblastic carcinoma (AC) is defined as a rare primary epithelial odontogenic malignant neoplasm and the malignant counterpart of benign epithelial odontogenic tumor of ameloblastoma (AB) by the WHO classification. AC develops pulmonary metastasis in about one third of the patients and reveals a poor prognosis. However, the mechanisms of AC oncogenesis remain unclear. In this report, we aimed to clarify the mechanisms of malignant transformation of AB or AC carcinogenesis. The relatively important genes in the malignant transformation of AB were screened by DNA microarray analysis, and the expression and localization of related proteins were examined by immunohistochemistry using samples of AB and secondary AC. Two genes of hypoxia-inducible factor 1 alpha subunit (HIF1A) and zinc finger E-box-binding homeobox 1 (ZEB1) were significantly and relatively upregulated in AC than in AB. Both genes were closely related in hypoxia and epithelial-mesenchymal transition (EMT). In addition, expressions of HIF-1α and ZEB1 proteins were significantly stronger in AC than in AB. In the cell assays using ameloblastoma cell line, AM-1, hypoxia condition upregulated the expression of transforming growth factor-β (TGF-β) and induced EMT. Furthermore, the hypoxia-induced morphological change and cell migration ability were inhibited by an antiallergic medicine tranilast. Finally, we concluded that hypoxia-induced HIF-1α and ZEB1 were critical for the malignant transformation of AB via TGF-β-dependent EMT. Then, both HIF-1α and ZEB1 could be potential biomarkers to predict the malignant transformation of AB.

Sun exposure and melanomas on sun-shielded and sun-exposed body areas.

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Malignant melanoma is a tumor that arises from melanocytes and accounts for around 4% of all malignancies in Europe and Northern America and for about 11% in Australia and New Zealand. About 10% of primary melanomas arise from sites not exposed to sun. Acral lentiginous melanoma, mucosal melanoma (in the oral cavities, nasal sinuses, genital tract and rectum) and uveal melanoma are all on non-sun-exposed tissues. Epidemiologic aspects ofmelanomas on non-sun-exposed areas in comparison with melanomas in sun-exposed areas have been reviewed. We focus on the relationship between melanoma incidence, geographic latitude of residence, race/ethnicity and host factors as well as time trends.

Comparison of the time and latitude trends of melanoma incidence in anorectal region and perianal skin with those of cutaneous malignant melanoma in Norway.

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Melanoma incidence is increasing in many parts of the world. The main environmental risk factor is exposure to solar radiation. However, melanomas may arise also on non-sun-exposed areas (uveal and mucosal melanomas) and little is known about a possible relationship between sun exposure and melanoma on such locations. We have compared the time and latitude trends of melanoma incidence in the anorectal region and perianal skin (non-sun-exposed sites) with those of cutaneous malignant melanoma (CMM) (sun-exposed skin) to gain more information about the relationship between sun exposure and melanoma on such sites. We analysed epidemiological data from the Cancer Registry of Norway for melanomas of the anorectal mucosa, perianal skin and overall CMM for the time period 1966-2007. We found that melanoma incidence on these shielded sites tends to decrease or remain constant over a period during which the CMM rates increase. This is true both in the North and in the South regions of Norway. Comparison of latitudinal trends of the incidence rates of CMM and melanoma on these shielded sites shows that there is no latitude gradient for melanoma of the anorectal mucosa and perianal skin, whereas there is a strong one for CMM. The time and latitudinal trends are likely to support the assumption that melanomas on these shielded sites are not generated by ultraviolet radiation. Possible causes and significances of these trends are discussed.

[Clinical and epidemiologic profile of melanoma patients according to sun exposure of the tumor site].

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Melanomas arising in areas with comparable levels of sun exposure have been shown to have similar genetic profiles. The aim of this study was to characterize the clinical features of melanoma patients according to the pattern of sun exposure: chronic, intermittent, or none. From our melanoma database, we selected 789 consecutive patients with melanoma diagnosed in our center since January 2000. Epidemiologic data, phenotype, and personal and family history of cancer were retrieved. The observed frequency of each variable was compared. Most melanoma patients presented tumors on areas exposed intermittently to sunlight. In addition, these patients presented higher numbers of common and atypical melanocytic nevi and the melanoma very frequently arose in a pre-existing nevus. The second largest group was formed by patients with tumors on areas chronically exposed to sun and that had all the clinical lesions (solar lentigines and actinic keratoses) and epidemiological characteristics typical of these areas. Finally, patients with melanomas on areas not exposed to sun were older, as occurred in the group with chronic exposure, and the diagnosis was made at more advanced stages of the disease. There are many patients who did not fit these patterns of melanoma development. Clinical and biological characterization is therefore necessary to determine alternative pathways of development in order to establish specific preventive measures.